

Information Hackathon

Prof. Dr. Dominik Heider

Machine Learning for Medical Data

Objective

Recent studies found new correlations between retina images and different biomarkers. These biomarkers can be used to identify the patient’s lifestyle or they can be indicators for diseases like dementia and cardiovascular disease. Most of the time, the necessary diagnostic information is derived by applying Machine Learning methods [1-3].

The goal of the Hackathon is to train a regression model to predict the chronological age. For healthy persons the difference between predicted age and real age should be small compared to the age difference for a person who is suffering from a disease [4].

The data is available here:

https://drive.google.com/drive/folders/1_SNpMzpDi5x8WRKbciGbTJyL3fy08mT7?usp=sharing

Outcome

We expect that each group tries to apply the techniques, that were taught in the lecture (this includes data preprocessing). The code that you write should have enough comments so that it is readable. You then have to submit your code, a README.txt file which shortly explains what you did and the conda environment. If the Code doesn’t run or has compatibility issues, the submission will be graded as failed.

In order to avoid misunderstandings, each group will come to our lab (only one time) for approximately one hour. In this time, you get to see the study design, code a minimal example for the Hackathon and you can participate in the study (so in the end you have your own data to put through your ML model).

What to do now

- 1 Form groups consisting of five students and choose a group name (be creative).
- 2 Open the AvailableDates.pdf and choose one available (blue) time slot. Make sure all team members have time and sent the chosen slot (data and time) together with the name of your team to Jan.Ruhland@hhu.de. He will update the available dates.
- 3 On the 13th of November the data set for the Hackathon will be published on Ilias, have a look together with your team.
- 4 Start to experiment with the data set and try your first machine learning models.
- 5 On the chosen date go to Graf-Adolf-Straße 63 (our lab is there).
- 6 Win the Hackathon!
- 7 If you have more questions come to the lecture on the 13th of November or use Ilias.

Contact Information

- Jan.Ruhland@hhu.de
- www.heiderlab.de

Schedule

- 1 13.11.23 - Information about Hackathon and working with images
- 2 15.11.23 - No lecture, start working on the data set
- 3 27.11.23 - No lecture, start of the Hackathon group meetings
- 4 29.11.23 - No lecture, Hackathon group meeting
- 5 04.12.23 - No lecture, Hackathon group meeting
- 6 06.12.23 - No lecture, Hackathon group meeting
- 7 14.01.23 - Leaderboard of top 3 teams will be published
- 8 29.01.23 - Deadline for submission (11.59pm)

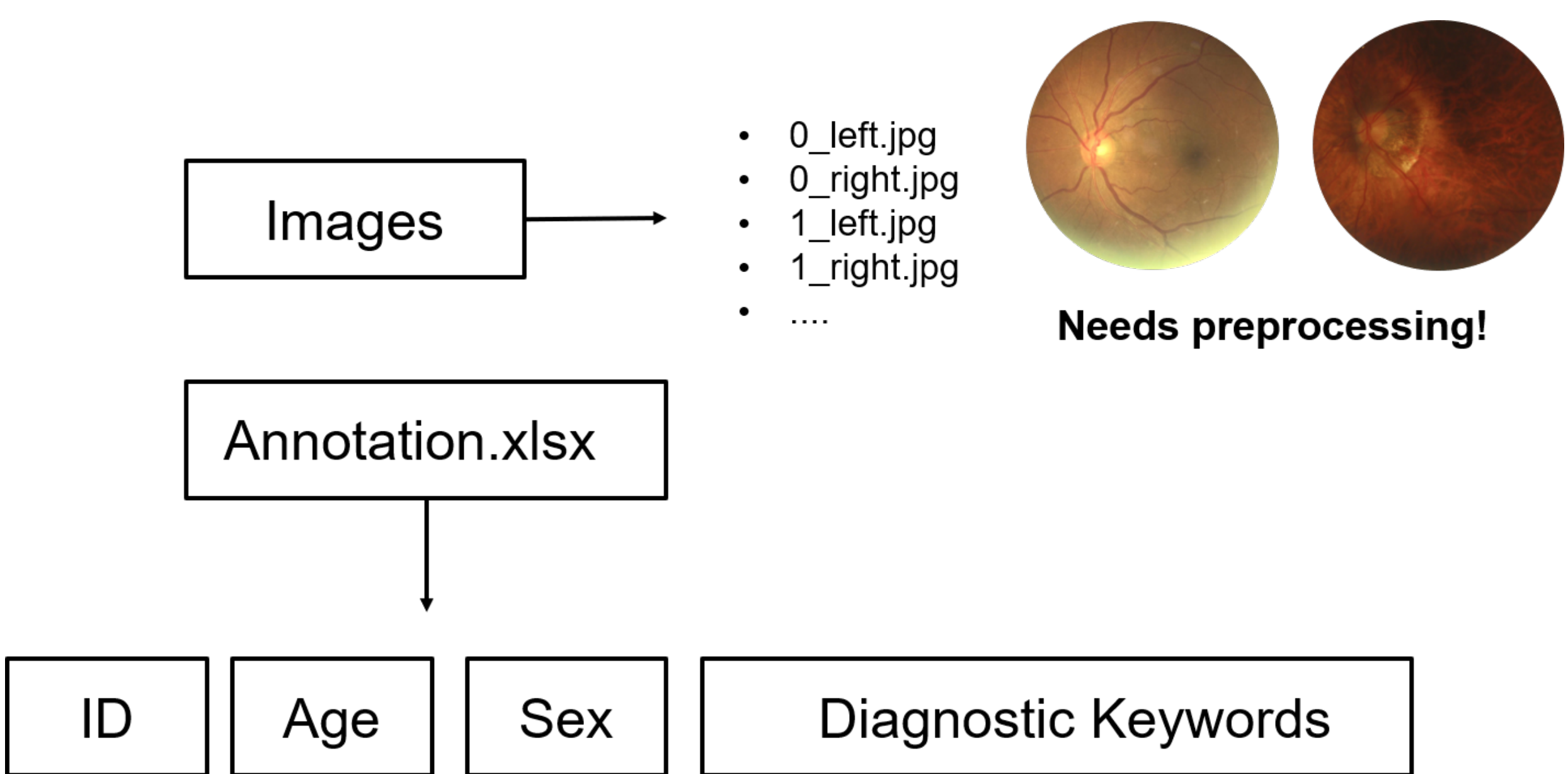


Figure 1: **Structure of the data set.** In the data folder you will find the folder "images" and the "Annotation.xlsx" file. The image folder contains all the retina images (from the left and the right eye) with the corresponding identification number. In the annotation file you will find all the information you need for the corresponding images.

References

- [1] Y. D. Kim, K. J. Noh, S. J. Byun, S. Lee, T. Kim, L. Sunwoo, K. J. Lee, S.-H. Kang, K. H. Park, and S. J. Park. Effects of hypertension, diabetes, and smoking on age and sex prediction from retinal fundus images. *Scientific Reports*, 10(1), Mar. 2020. doi: 10.1038/s41598-020-61519-9. URL <https://doi.org/10.1038/s41598-020-61519-9>.
- [2] S. McGrory, J. Cameron, E. Pellegrini, C. Warren, F. Doubal, I. Deary, B. Dhillon, J. Wardlaw, E. Trucco, and T. MacGillivray. The application of retinal fundus camera imaging in dementia: A systematic review. *Alzheimer’s Dementia: Diagnosis, Assessment Disease Monitoring*, 6:91–107, 2017. doi: 10.1016/j.dadm.2016.11.001. This work was supported by the Engineering and Physical Sciences Research Council (EPSRC) grant “Multi-modal retinal biomarkers for vascular dementia” (EP/M005976/1).
- [3] M. R. Munk, T. Kurmann, P. Márquez-Neila, M. S. Zinkernagel, S. Wolf, and R. Sznitman. Assessment of patient specific information in the wild on fundus photography and optical coherence tomography. *Scientific Reports*, 11(1), Apr. 2021. doi: 10.1038/s41598-021-86577-5. URL <https://doi.org/10.1038/s41598-021-86577-5>.
- [4] A. Zhavoronkov and P. Mamoshina. Deep aging clocks: The emergence of AI-based biomarkers of aging and longevity. *Trends in Pharmacological Sciences*, 40(8):546–549, Aug. 2019. doi: 10.1016/j.tips.2019.05.004. URL <https://doi.org/10.1016/j.tips.2019.05.004>.